

Sobha Realty

Delayed Transaction & Revenue Exposure Analysis

Milestone 1

Group Number: 08

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1. Problem Statement

Sobha Realty has a large portfolio of property transactions across multiple projects and property types. A significant number of transactions in the available 2018-2023 dataset are classified as delayed. This project analyzes where these delayed transactions are concentrated across projects, property types, and years, quantifies the revenue associated with them, and compares delay rates across projects to identify areas that may warrant greater management attention. The analysis is limited to patterns observable in the transaction data and does not attempt to determine the underlying causes of delays.

A transaction is treated as delayed when its ProcedureNameEn field carries one of the following status labels: Complete Delayed Sell, Delayed Sell, Delayed Mortgage, Grant on Delayed Sell, Delayed Sell Lease to Own Registration.

2. Dataset

The dataset was obtained from an actual client business (identity kept confidential for privacy). It originally had 105,283 rows covering property transactions from 2018 to 2023 across 116 projects.

Before analysis, the data was cleaned in Excel:

- Duplicate rows were removed
- Rows with missing or invalid project names were removed
- Rows with a price of 0 or blank were removed
- Columns with personal buyer/seller information (name, mobile number, ID numbers) were deleted for privacy reasons

After cleaning, the analysis was run only on rows where the party type is 'Buyer', to avoid counting the same transaction twice (each transaction otherwise appears once for the buyer and once for the seller).

3. Data Dictionary

The table below explains the columns used in this analysis, along with data quality notes found during inspection.

Column	Meaning	Data Type	Example	Quality Notes
Regis	Date the transaction was registered	Date (mixed text formats)	28/12/2023	Two different date formats found in the raw file; handled using TRY_STRPTIME in SQL
Project	Name of the real estate project	Text	Sobha Hartland Waves	Some rows had '0' as a placeholder; these rows were removed during cleaning
PropertyTypeEn	Type of property (Flat, Villa, Unit, etc.)	Text	Flat	A few rows are blank/null
ProcedureValue	Transaction value in AED	Stored as text in the CSV	1700000	Converted using TRY_CAST(...AS DOUBLE) in SQL; rows with 0 or blank value were removed
ProcedureNameEn	Transaction status/type	Text	Delayed Sell	Used to identify delayed transactions (see list below)
ProcedurePartyTypeNameEn	Whether the row is the Buyer or Seller side of the transaction	Text	Buyer	Analysis is filtered to 'Buyer' only, to avoid double-counting each transaction
Size	Property size	Numeric	1250	Not used directly in this milestone's queries

Column	Meaning	Data Type	Example	Quality Notes
BuildingNameEn	Building name within a project	Text	Building 3	Not used directly in this milestone's queries
CountryNameEn	Nationality of buyer/seller	Text	India	Not used directly in this milestone's queries

4. Tools Used

- SQL - DataLab (DataCamp), using DuckDB
- Microsoft Excel - data cleaning and preparation

5. Questions Answered

- How many total buyer-level transactions are there?
- What percentage of all transactions are delayed?
- How many delayed transactions happened in each year from 2018 to 2023?
- Which projects have the highest number of delayed transactions?
- Which property type is delayed the most?
- Is the average price of a delayed transaction higher or lower than a normal one?
- Which projects lose the biggest share of their revenue to delayed transactions?

6. SQL Techniques Used

SELECT, WHERE, filtering, aggregation (COUNT, SUM, AVG), GROUP BY, ORDER BY, and basic cleaning were used across the queries below. CASE WHEN was also used in most queries, but only to identify whether a transaction is delayed based on the ProcedureNameEn field, since the dataset does not have a single 'is_delayed' column. It was not used to create new business categories, which is planned for Milestone 2. TRY_CAST and TRY_STRPTIME were used to safely convert text values (price and date fields) that were not stored in a clean numeric or date format in the source file.

7. Analysis

Q1. How many total buyer-level transactions are there?

Why this query: establishes the overall transaction volume used as the base for every other calculation.

```
SELECT COUNT(*) AS total_transactions
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer';
```

Output: total_transactions = 46,763

Q2. What percentage of all transactions are delayed?

Why this query: calculates how many of the total transactions are delayed, and what share of the total that represents.

```
SELECT
COUNT(*) AS total_transactions,
SUM(CASE WHEN ProcedureNameEn IN
('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
THEN 1 ELSE 0 END) AS delayed_transactions,
ROUND(
SUM(CASE WHEN ProcedureNameEn IN
```

```

('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
) AS delay_percentage
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer';

```

Output: total_transactions = 46,763 | delayed_transactions = 12,683 | delay_percentage = 27.12%

Q3. How many delayed transactions happened in each year from 2018 to 2023?

Why this query: groups transactions by year to see whether delays are getting better or worse over time, not just what the current overall rate is. The Regis column contained two different date formats in the raw file, so TRY_STRPTIME with two format patterns was used to read it correctly.

```

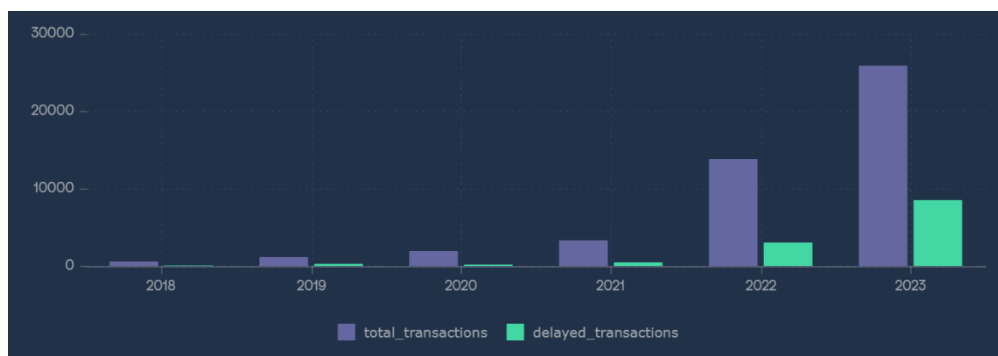
SELECT
  YEAR(
    COALESCE(
      TRY_STRPTIME(Regis, '%d/%m/%Y'),
      TRY_STRPTIME(Regis, '%d-%b-%y')
    )
  ) AS transaction_year,
  COUNT(*) AS total_transactions,
  SUM(CASE WHEN ProcedureNameEn IN
    ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
    'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
    THEN 1 ELSE 0 END) AS delayed_transactions
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'
GROUP BY 1
ORDER BY 1;

```

Year	Total Transactions	Delayed Transactions
2018	594	68
2019	1,178	277
2020	1,951	215
2021	3,349	470
2022	13,847	3,085
2023	25,844	8,568

Chart 1 (for Q3) - Year-wise total vs. delayed transactions, 2018-2023

Total transaction volume grew sharply over the period, from 594 transactions in 2018 to 25,844 in 2023. Delays grew even faster in relative terms: only about 11% of 2018 transactions were delayed, versus about 33% in 2023. This suggests that as Sobha's sales volume scaled up quickly, the company's ability to close out transactions on time did



not keep pace.

Q4. Which projects have the highest number of delayed transactions?

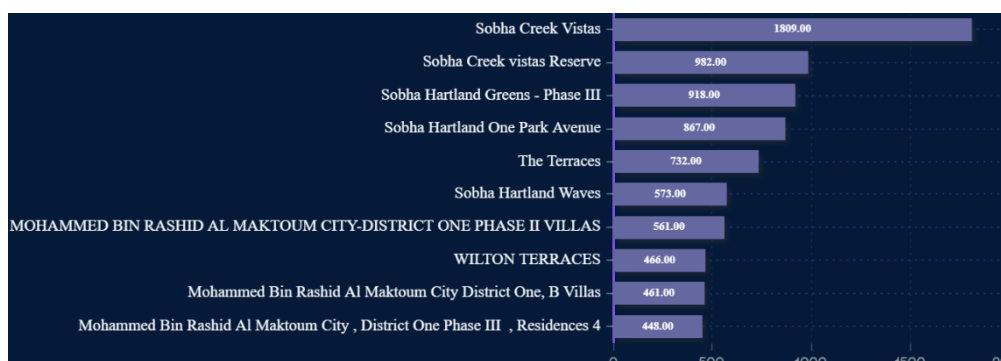
Why this query: ranks projects by raw delayed-transaction count to identify where management attention is most urgently needed in absolute terms.

```
SELECT
  Project,
  COUNT(*) AS delayed_count
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'
AND ProcedureNameEn IN
  ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
   'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
GROUP BY Project
ORDER BY delayed_count DESC
LIMIT 10;
```

Project	Delayed Transactions
Sobha Creek Vistas	1,809
Sobha Creek Vistas Reserve	982
Sobha Hartland Greens - Phase III	918
Sobha Hartland One Park Avenue	867
The Terraces	732
Sobha Hartland Waves	573
Mohammed Bin Rashid Al Maktoum City - District One Phase II Villas	561
Wilton Terraces	466
Mohammed Bin Rashid Al Maktoum City, District One, B Villas	461
Mohammed Bin Rashid Al Maktoum City, District One Phase III, Residences 4	448

Chart 2 (for Q4) - Top 10 projects by number of delayed transactions

Sobha Creek Vistas has almost double the delayed transactions of the next closest project, and alone accounts for around AED 1.59 billion tied up in delayed transactions.



Q5. Which property type is delayed the most?

Why this query: breaks delays down by property type, and calculates delay_percentage (not just raw count) so that small but high-risk categories are not hidden behind larger ones.

```
SELECT
  PropertyTypeEn,
  COUNT(*) AS total_transactions,
  SUM(CASE WHEN ProcedureNameEn IN
    ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
```

```

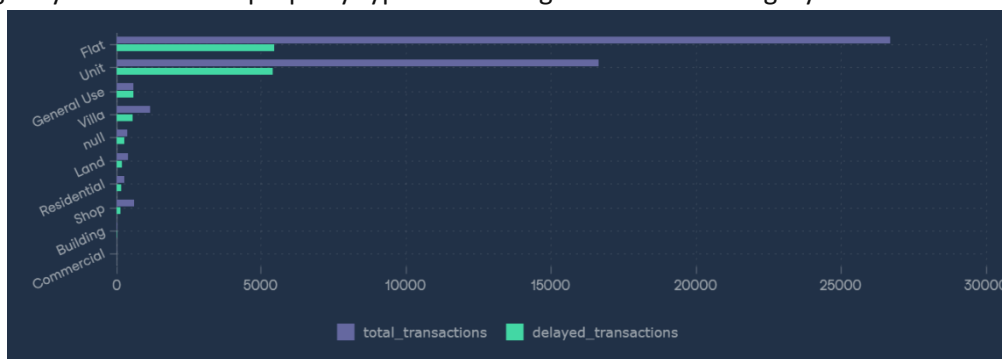
'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
THEN 1 ELSE 0 END) AS delayed_transactions,
ROUND(
SUM(CASE WHEN ProcedureNameEn IN
('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
) AS delay_percentage
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'
GROUP BY PropertyTypeEn
ORDER BY delayed_transactions DESC;

```

Property Type	Total Transactions	Delayed Transactions
Flat	26,699	5,435
Unit	16,632	5,387
General Use	569	567
Villa	1,164	540
(blank/null)	378	253
Land	407	185
Residential	274	166
Shop	616	136
Building	19	14
Commercial	5	0

Chart 3 (for Q5) - Total vs. delayed transactions by property type

By raw count, Flats and Units have the most delayed transactions simply because they are by far the largest categories. But by percentage, General Use properties stand out: 567 of 569 General Use transactions (99.6%) are delayed, making it by far the riskiest property type even though it is a small category.



Q6. Is the average price of a delayed transaction higher or lower than a normal one?

Why this query: compares average transaction value for delayed vs. normal transactions to see whether higher-value deals are more or less likely to be delayed. ProcedureValue is stored as text in the file, so TRY_CAST is used to safely convert it to a number.

```

SELECT
CASE WHEN ProcedureNameEn IN
('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
THEN 'Delayed' ELSE 'Normal' END AS transaction_status,
ROUND(AVG(TRY_CAST(ProcedureValue AS DOUBLE)), 2) AS avg_price,
COUNT(*) AS total_count
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'
AND TRY_CAST(ProcedureValue AS DOUBLE) > 0
GROUP BY transaction_status;

```

Output: avg_delayed_price = AED 2,011,759.92 | avg_normal_price = AED 2,422,768.50

Delayed transactions have a lower average price than normal transactions. This suggests that lower-value properties are somewhat more likely to be delayed than higher-value ones.

Q7. Which projects lose the biggest share of their revenue to delayed transactions?

Why this query: looks at revenue exposure rather than transaction count, and filters out very low-volume projects (fewer than 30 transactions) so a single delayed sale in a tiny project doesn't distort the ranking.

```
SELECT
  Project,
  SUM(TRY_CAST(ProcedureValue AS DOUBLE)) AS total_project_value,
  SUM(CASE WHEN ProcedureNameEn IN
    ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
    'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
    THEN TRY_CAST(ProcedureValue AS DOUBLE) ELSE 0 END) AS delayed_value,
  ROUND(
    SUM(CASE WHEN ProcedureNameEn IN
    ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
    'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
    THEN TRY_CAST(ProcedureValue AS DOUBLE) ELSE 0 END) * 100.0
    / SUM(TRY_CAST(ProcedureValue AS DOUBLE)), 2
  ) AS pct_value_delayed
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer' AND TRY_CAST(ProcedureValue AS DOUBLE) > 0
GROUP BY Project
HAVING COUNT(*) >= 30
ORDER BY pct_value_delayed DESC
LIMIT 10;
```

Output: Four projects under Mohammed Bin Rashid Al Maktoum City have 100% of their recorded revenue sitting in delayed transactions. Sobha Creek Vistas has 77.56% of its revenue (about AED 1.59 billion) in delayed transactions.

This shows that sales volume alone is not enough to judge project health - a project can sell steadily while almost none of that revenue is actually confirmed as on-time.

8. Key Findings

1. Close to 1 in 4 transactions are delayed

Out of 46,763 buyer-level transactions, 12,683 are delayed - 27.12%. Roughly one out of every four transactions ends up delayed, which shows this is not a small or occasional problem.

2. Delays have been getting worse over time, especially in 2023

In 2018, only about 11% of transactions were delayed. By 2023, that number had climbed to around 33%. Over the same period, total transaction volume grew from 594 to 25,844. As Sobha's sales grew quickly, the company may have struggled to keep up, which shows up as more delays.

3. Sobha Creek Vistas has the most delayed transactions

Sobha Creek Vistas has 1,809 delayed transactions, almost double the next highest project. In money terms, this single project has around AED 1.59 billion tied up in delayed transactions.

4. "General Use" properties are delayed almost every time

By raw count, Flats and Units appear to have the most delays. But by percentage, General Use stands out: 567 of 569 of its transactions (99.6%) are delayed. It's a small category, but by far the riskiest - showing that percentages, not just raw counts, give a more accurate picture.

5. Some projects have their entire revenue tied up in delays

Four projects under Mohammed Bin Rashid Al Maktoum City have 100% of their revenue sitting in delayed transactions. This means it isn't enough to look at which project sells the most - it also matters whether that project's money is actually coming in on time.

9. Limitations

This analysis is based only on the transaction status field in the dataset, which labels certain transactions as delayed. The dataset does not include an expected or actual handover date, so this project cannot say why a transaction was delayed or confirm that it reflects a late property handover to the buyer. The findings only describe patterns in the recorded transaction data.

10. What We Learned

This project helped us understand how to move from a large, messy dataset to a clear business question, and how to use SQL to answer it step by step. We also learned that raw counts alone can be misleading, and that calculating percentages and revenue exposure gives a much clearer picture of where the real risk is.